

Executive Summary

Hospital readmissions remain a significant challenge for healthcare systems, impacting patient outcomes, hospital capacity, and overall costs. Reducing preventable readmissions is a key priority, particularly in value-based care models where hospitals are evaluated on quality and efficiency.

This project analyzes a large dataset of hospital encounters to identify patterns associated with patient readmissions and to develop a predictive model that can flag high-risk patients. The dataset includes over 100,000 records with variables related to patient demographics, clinical conditions, and prior healthcare utilization.

Two classification models were developed and evaluated: Logistic Regression and Random Forest. While Logistic Regression achieved a high overall accuracy of 0.89, it performed poorly in identifying readmitted patients, with a recall of only 0.02. In contrast, the Random Forest model achieved a lower accuracy of 0.65 but significantly improved recall of 0.56, making it more effective for identifying patients at risk of readmission.

The results demonstrate that accuracy alone is not an appropriate metric for this problem. From a business and clinical perspective, the ability to correctly identify high-risk patients is more valuable than overall accuracy. The Random Forest model provides actionable insights that can support targeted interventions, improve care coordination, and reduce avoidable readmissions.

Introduction & Business Context

Hospital readmissions are widely used as an indicator of healthcare quality and system performance. High readmission rates may reflect gaps in discharge planning, care coordination, or follow-up care. In addition to negatively impacting patient outcomes, readmissions place a financial burden on hospitals through increased utilization and potential penalties under value-based reimbursement models.

The objective of this project is to develop a predictive model that identifies patients at risk of hospital readmission. By proactively identifying these individuals, healthcare providers can implement targeted interventions, such as enhanced discharge planning or follow-up care, to reduce the likelihood of readmission.

This problem is particularly relevant in today's healthcare environment, where organizations are increasingly focused on improving outcomes while controlling costs. Predictive analytics enables hospitals to shift from reactive care to proactive, data-driven decision-making.

The analysis is guided by the following research questions:

- Which patient and clinical factors are most strongly associated with hospital readmissions?
- How effectively can readmissions be predicted using available data?
- Which modeling approach provides the most practical value in a healthcare setting?

The dataset used in this analysis includes patient-level information such as demographics, diagnoses, medications, and prior healthcare utilization. The dataset includes over 100,000 hospital encounters across multiple facilities in the United States. These variables provide a comprehensive view of both clinical complexity and patient behavior, which are critical drivers of readmission risk.

Exploratory Data Analysis

The exploratory data analysis focused on understanding the dataset, identifying patterns related to hospital readmissions, and examining relationships between key variables.

The dataset includes over 100,000 hospital encounters with a mix of demographic, clinical, and utilization-based variables. Initial analysis revealed a clear imbalance in the target variable, with significantly more patients not readmitted than readmitted. This imbalance is important because it affects model performance and makes accuracy alone an unreliable metric.

Readmission Distribution

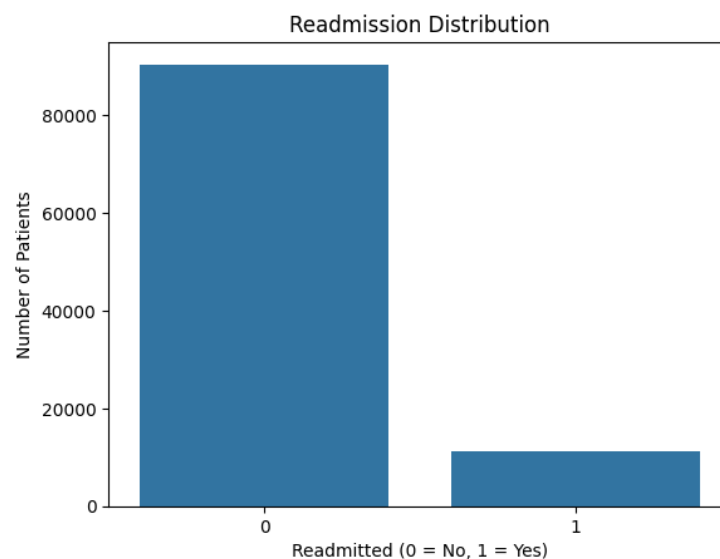


Figure 1: Readmission Distribution

The distribution shows a clear class imbalance, with significantly more patients not readmitted

than readmitted within 30 days. This imbalance highlights the need to use evaluation metrics beyond accuracy, such as recall, when assessing model performance.

Length of Stay and Readmission

To understand clinical factors, the relationship between hospital stay duration and readmission was examined. Patients with longer hospital stays tend to have higher readmission rates, suggesting that length of stay reflects clinical complexity.

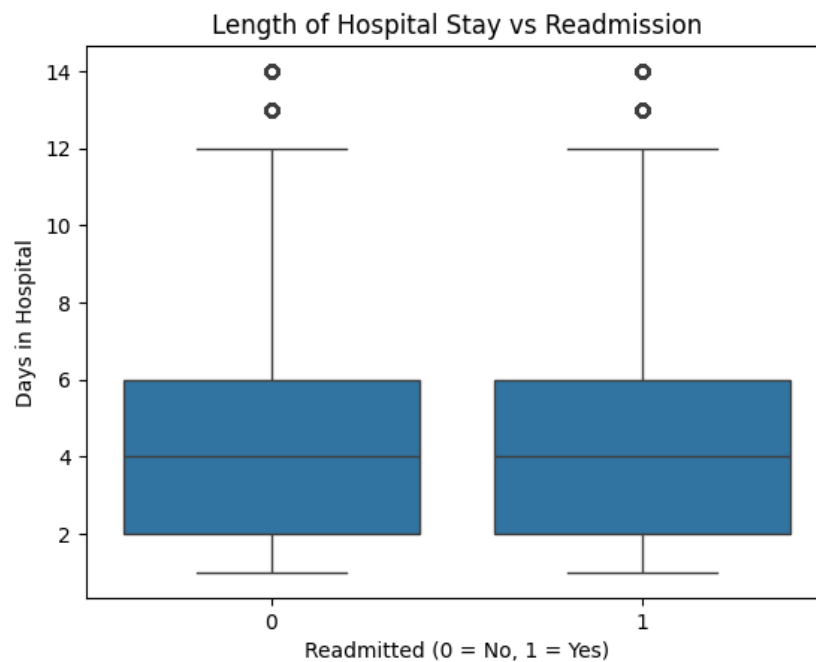


Figure 2: Time in Hospital vs Readmission

Patients with longer hospital stays show a higher likelihood of readmission.

Prior Utilization and Readmission

Prior healthcare utilization is a strong indicator of future readmissions. Patients with more inpatient visits were more likely to be readmitted, highlighting the importance of historical utilization patterns.

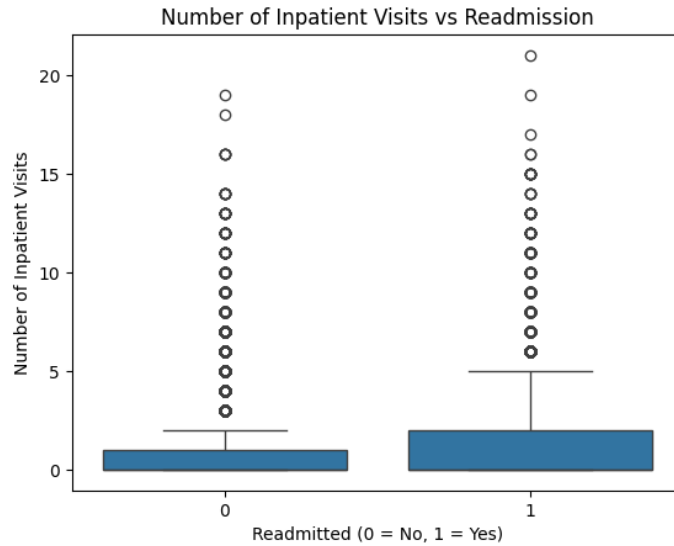


Figure 3: Inpatient Visits vs Readmission

Patients with more prior inpatient visits are more likely to be readmitted.

Clinical Intensity and Readmission

Clinical intensity was analyzed using the number of medications. Patients with higher medication counts showed increased readmission risk, likely reflecting more complex health conditions.

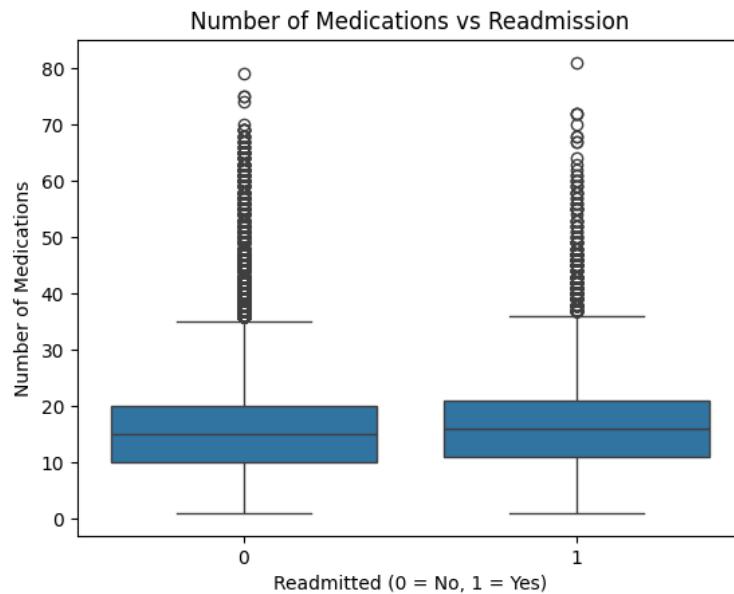


Figure 4: Number of Medications vs Readmission

Higher medication counts are associated with increased readmission risk.

Demographics and Readmission

Demographic patterns, such as age, were also explored. While differences exist across age groups, these patterns are less pronounced compared to clinical and utilization factors.

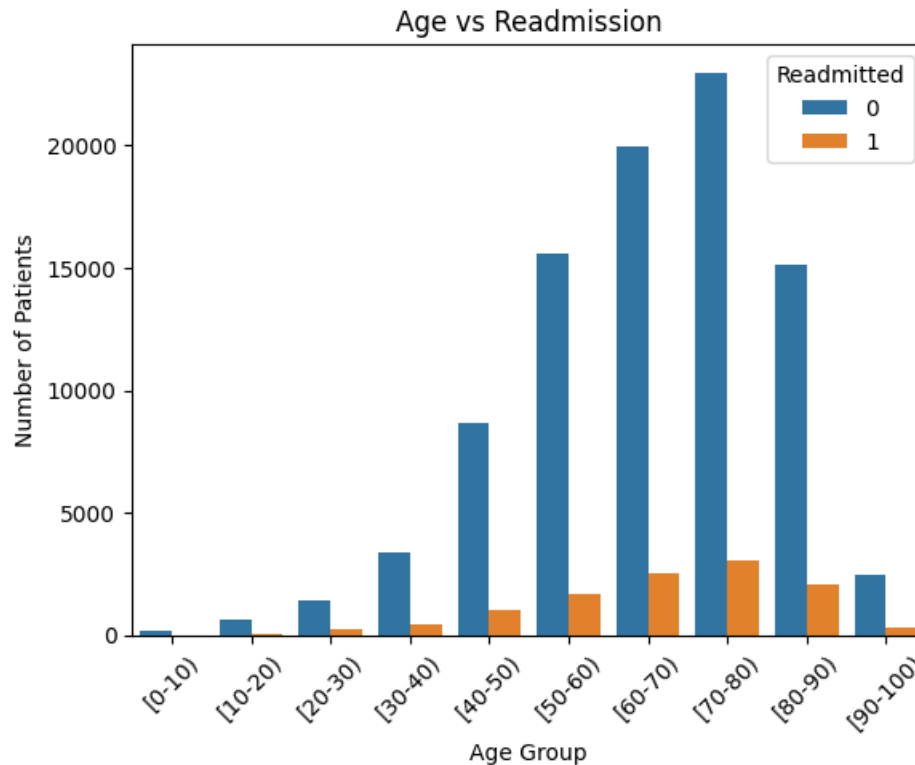


Figure 5: Age vs Readmission

Readmission rates vary across age groups, though clinical factors appear to have a stronger impact.

Summary of Findings

Overall, the exploratory analysis shows that hospital readmissions are primarily driven by prior healthcare utilization and clinical complexity. Variables such as length of stay, number of inpatient visits, and number of medications are key indicators of readmission risk.

These findings informed the modeling approach and supported the decision to prioritize evaluation metrics such as recall in later stages of the analysis.

Methodology

A structured approach was used to prepare the data and develop predictive models for hospital readmissions.

Data Preprocessing

The dataset was first cleaned to ensure consistency and usability. Missing values represented by “?” were replaced with null values, and variables with a high proportion of missing data, such as weight, payer code, and medical specialty, were removed. Identifier columns, including encounter ID and patient number, were also dropped as they do not contribute to prediction.

A new binary target variable, `readmitted_binary`, was created to simplify the prediction task. Patients readmitted within 30 days were labeled as 1, while all others were labeled as 0.

Categorical variables were encoded to allow them to be used in machine learning models, and the dataset was prepared for modeling by separating features and the target variable.

Model Selection

Two classification models were selected:

- **Logistic Regression** was used as a baseline model due to its simplicity and interpretability
- **Random Forest** was used to capture more complex relationships and interactions between variables

These models were chosen to compare a simple, interpretable approach with a more flexible, higher-performing method.

Model Training

The dataset was split into training and testing sets to evaluate model performance on unseen data. Each model was trained using the training dataset and then evaluated on the test dataset.

The Random Forest model was configured with multiple trees and included class balancing to account for the imbalance in readmission outcomes. The data was split into training and testing sets (e.g., 80/20 split) to ensure unbiased model evaluation.

Evaluation Metrics

Model performance was evaluated using the following metrics:

- **Accuracy** to measure overall correctness
- **Precision** to evaluate how many predicted readmissions were correct
- **Recall** to measure how well the model identified actual readmissions
- **F1-score** to balance precision and recall

Given the class imbalance and the importance of identifying high-risk patients, recall was the most important metric in this analysis.

Additional Insights from EDA

The additional visualizations further reinforce the importance of clinical and utilization variables in predicting readmissions.

Patients with a higher number of inpatient visits show a clear increase in readmission likelihood, indicating that prior hospital usage is a strong predictor of future readmissions. Similarly, patients taking a greater number of medications tend to have higher readmission rates, which reflects increased clinical complexity.

The analysis of length of hospital stay also supports this pattern, as patients with longer stays are more likely to be readmitted. This suggests that more severe or complex cases are associated with higher risk.

While age shows some variation in readmission rates, it appears to be less influential compared to utilization and clinical intensity variables.

Results & Model Comparison

Model Performance Summary

Model	Accuracy	Precision (Class 1)	Recall (Class 1)	F1-Score (Class 1)
Logistic Regression	0.89	0.42	0.02	0.03
Random Forest	0.65	0.17	0.56	0.26

Key Findings

The Logistic Regression model achieved high overall accuracy of 0.89. However, it performed poorly in identifying patients who were actually readmitted, with a recall of only 0.02. This indicates that the model is heavily biased toward predicting patients as not readmitted, making it ineffective for identifying high-risk individuals.

In contrast, the Random Forest model achieved a lower accuracy of 0.65 but significantly improved recall of 0.56. This means the model successfully identified over half of the patients who were readmitted, making it far more useful in practice.

Best Model Selection

The Random Forest model was selected as the best-performing model for this analysis.

While it has lower overall accuracy, it is much more effective at identifying patients at risk of readmission. In a healthcare setting, correctly identifying high-risk patients is more important than overall accuracy, as missed readmissions can lead to worse outcomes and higher costs.

Feature Importance

To better understand what drives predictions, feature importance from the Random Forest model was analyzed.

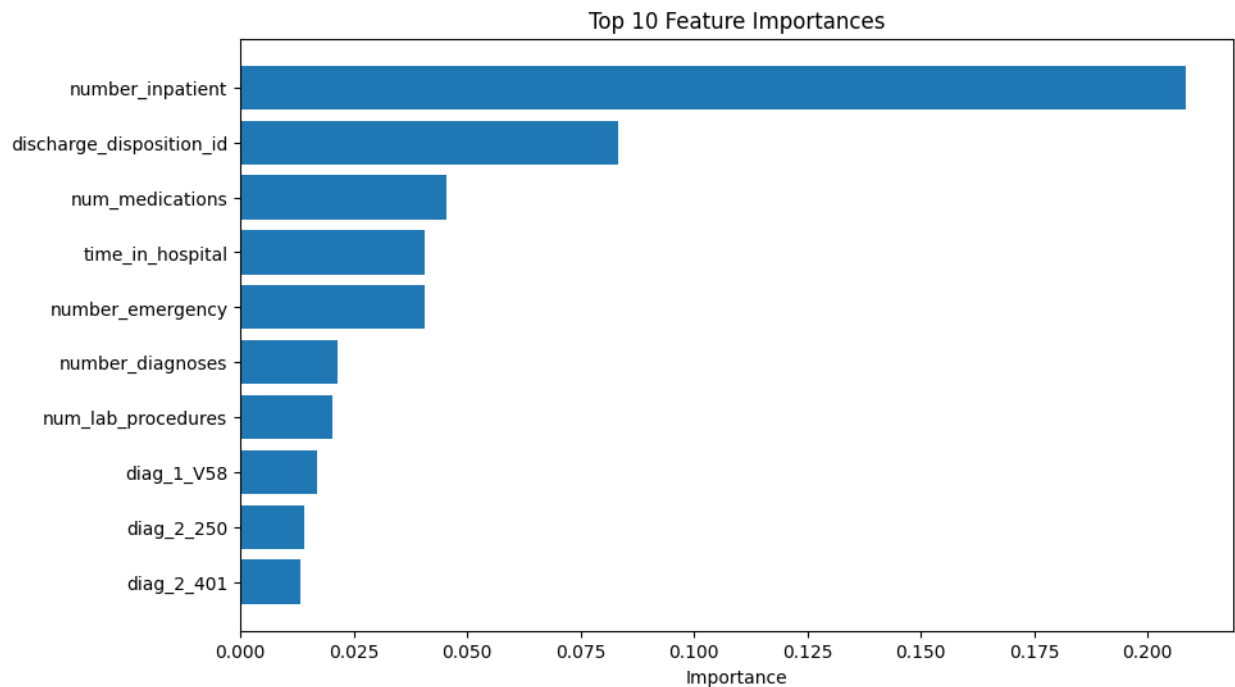


Figure 6: Top Predictors of Hospital Readmission

Prior inpatient visits, discharge disposition, and clinical intensity variables are the strongest predictors of readmission risk.

Feature importance analysis shows that **number of inpatient visits** is the most significant predictor of readmission. This reinforces the importance of prior healthcare utilization as a key driver. Other important variables include discharge disposition, number of medications, and length of hospital stay, all of which reflect patient complexity and care needs.

Summary of Results

Overall, the results demonstrate that model selection should be based on practical usefulness rather than accuracy alone. The Random Forest model provides meaningful predictive value by identifying a substantial portion of high-risk patients, making it a strong candidate for real-world implementation.

Business Insights & Recommendations

The results of this analysis provide clear opportunities for improving patient care and hospital operations.

The Random Forest model identified that patients with higher prior healthcare utilization, longer hospital stays, and greater clinical complexity are at increased risk of readmission. These insights align with real-world healthcare patterns, where patients with more frequent interactions with the healthcare system often require additional support after discharge.

From a business perspective, these findings can be used to improve both patient outcomes and operational efficiency. Hospitals can implement a predictive approach to identify high-risk patients before discharge and take targeted action.

Recommendations

Hospitals should use predictive models to flag patients at high risk of readmission at the time of discharge. This allows care teams to prioritize follow-up appointments, ensure proper discharge planning, and provide additional resources such as care coordination or patient education.

Patients identified as high risk should receive more intensive post-discharge support, including follow-up calls, medication management, and scheduling of early outpatient visits. These interventions can help prevent avoidable readmissions.

Additionally, hospitals can use these insights to allocate resources more effectively. Rather than applying the same level of care to all patients, resources can be focused on those who need it most, improving efficiency while maintaining quality of care.

Expected Impact

Implementing this approach can lead to reduced readmission rates, improved patient outcomes, and lower costs associated with unnecessary hospital visits. It also supports hospitals in meeting quality benchmarks and avoiding penalties under value-based care models.

Ethics & Responsible AI

The use of predictive models in healthcare introduces important ethical considerations that must be addressed to ensure fair and responsible use.

One key concern is the potential for bias in the dataset. If certain patient populations are underrepresented or have historically received different levels of care, the model may not perform equally well across all groups. This could lead to unequal identification of high-risk patients and disparities in care.

Fairness should be considered by evaluating how the model performs across different demographic groups, such as age or other available patient characteristics. Ensuring that the model does not disproportionately overlook certain populations is critical.

Patient privacy is another important consideration. Healthcare data is highly sensitive, and all data used in this analysis must be handled in compliance with privacy regulations such as HIPAA. Any real-world implementation of this model would require secure data storage and strict access controls.

To support responsible use, the model should be regularly monitored and updated to ensure continued accuracy and fairness. Transparency is also important, so healthcare providers understand how predictions are being used to support decision-making.

Conclusion & Future Work

This project demonstrates how predictive analytics can be applied to identify patients at risk of hospital readmission and support proactive healthcare decision-making.

While Logistic Regression achieved high overall accuracy, it failed to effectively identify readmitted patients, making it less useful in practice. The Random Forest model, despite lower accuracy, provided significantly better recall and was more effective in identifying high-risk individuals.

The results highlight the importance of selecting evaluation metrics that align with real-world objectives. In this case, recall was more meaningful than accuracy because identifying at-risk patients is critical for improving outcomes.

There are several opportunities for future improvement. Additional data, such as social determinants of health, could improve model performance and provide a more complete view of patient risk. More advanced modeling techniques could also be explored to further enhance predictive accuracy.

Overall, this project shows that data-driven approaches can support more proactive and effective healthcare decision-making.

References & Acknowledgments

Dataset:

Diabetes 130-US hospitals dataset, UCI Machine Learning Repository

Tools and Technologies:

Python, Google Colab, GitHub

Libraries Used:

pandas, numpy, scikit-learn, matplotlib, seaborn